

PROJECT REPORT

1. INTRODUCTION

1.1 Project Overview

In today's digital era, effectively managing personal finances is essential, yet often challenging amidst complex tools. Many existing financial apps can overwhelm users with cluttered interfaces or unnecessary features, detracting from the core goal of financial clarity. The **MERN Personal Expense Tracker** emerges as a user-friendly and comprehensive solution designed to simplify this process. Built for individuals, working professionals, students, and families alike, this application enables users to effortlessly track expenses, categorize transactions, establish and monitor budgets across various periods, and receive timely alerts. The project was developed following a user-centered design approach — covering every stage from user research and pain point analysis to ideation, prototyping, visual design, and usability testing. The focus was to bridge the gap between user needs and practical design solutions with simplicity at the core.

1.2 Purpose

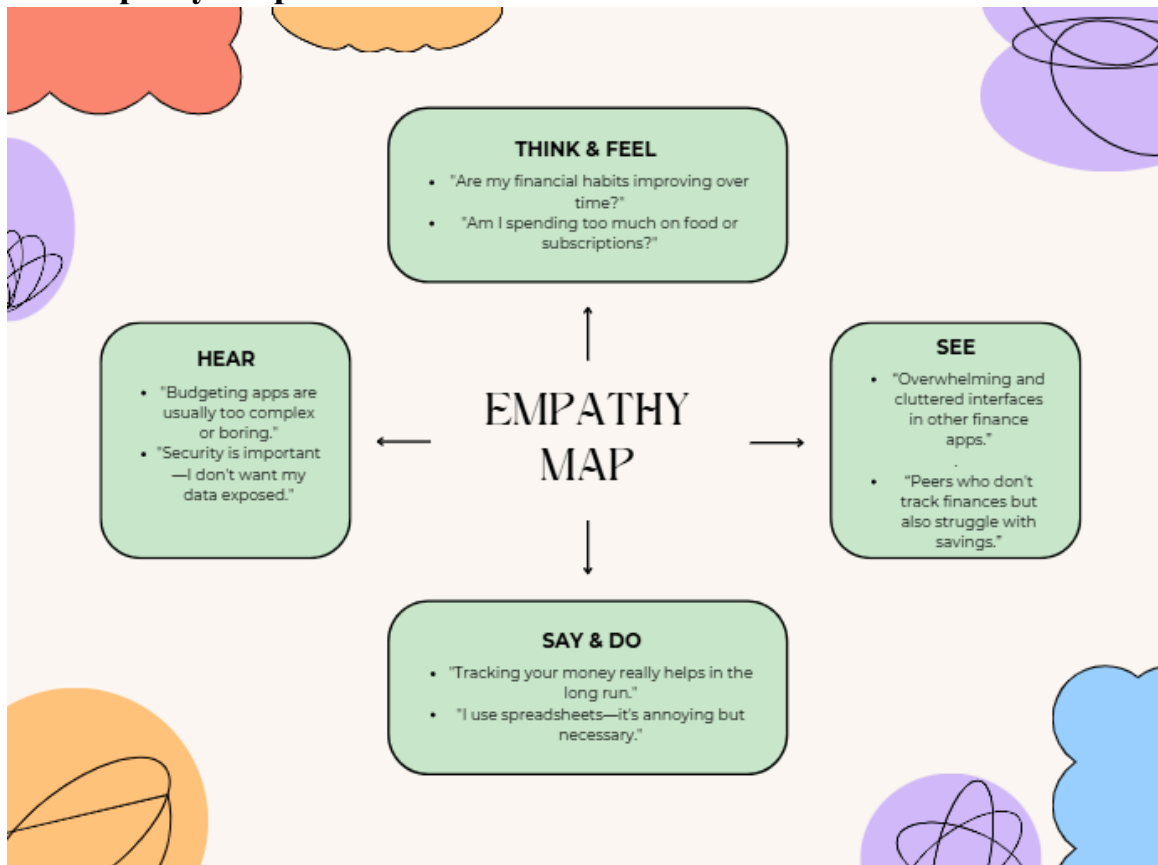
By providing clear visual reports on spending patterns and budget adherence (spending by category, budget vs. actuals), the tracker aims to deliver actionable insights. This empowers users to gain a firm grasp on their cash flow and make informed decisions towards achieving their financial goals. Key features include secure user authentication, full CRUD operations for expenses and budgets, dynamic reporting, category management, budget progress visualization, and a notification system for budget limits. The development process focused on integrating these essential financial management tools within an intuitive, animated, and responsive interface, prioritizing clarity and user control.

2. IDEATION PHASE

2.1 Problem Statement

User Type	Wants To	But	Because
College Student	Manage limited money	Struggles to track expenses	Tools are complex or not beginner-friendly
Young Professional	Stay on budget	Hard to stick to a plan	Lacks real-time insights or motivation
Teenage Student	Control spending	Lacks guidance	No exposure to finance tools
Neurodivergent User	Build spending habits	Gets overwhelmed by apps	Apps lack accessibility features
Budgeting Enthusiast	Save more effectively	Needs deeper insights	Tools aren't customizable enough

2.2 Empathy Map Canvas



2.3 Brainstorming

The Anxious Beginner:

"I'm scared to look at my finances. Will this app just make me feel worse?"

"I don't even know where to start with budgeting. Is this app going to be too complicated?"

"I wish someone would just tell me what to do with my money."

The Busy Professional:

"I don't have time to manually track every expense. Will this app automate things for me?"

"I need to see the big picture quickly. Can this app provide a clear overview of my finances?"

"I hope this app will give me insights I wouldn't have seen on my own."

The Frugal Student:

"I need to track every penny. Can this app handle detailed expense tracking?"

"I want to start good financial habits now. Will this app help me learn?"

"I hope this app can help me save for school."

The Older Adult:

"I'm not very tech-savvy. Is this app easy to use?"

"I'm focused on retirement planning. Can this app help me with that?"

"I hope this app will simplify my finances."

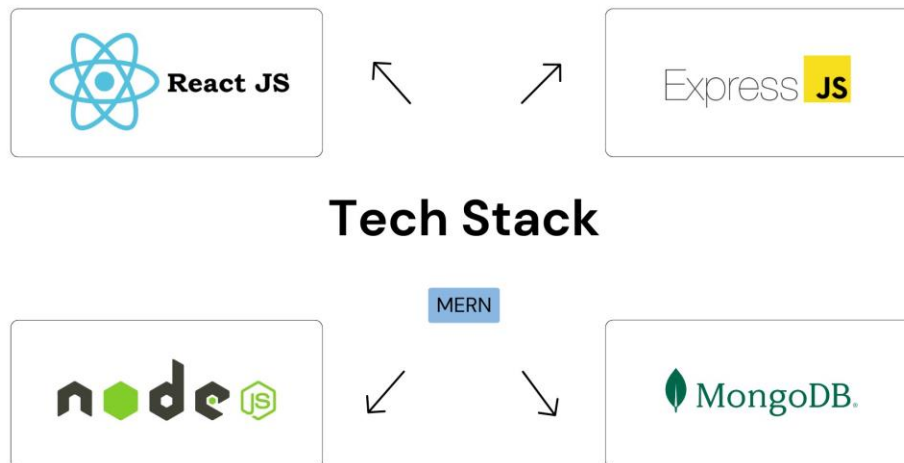
3. REQUIREMENT ANALYSIS

Mapping the Customer Journey

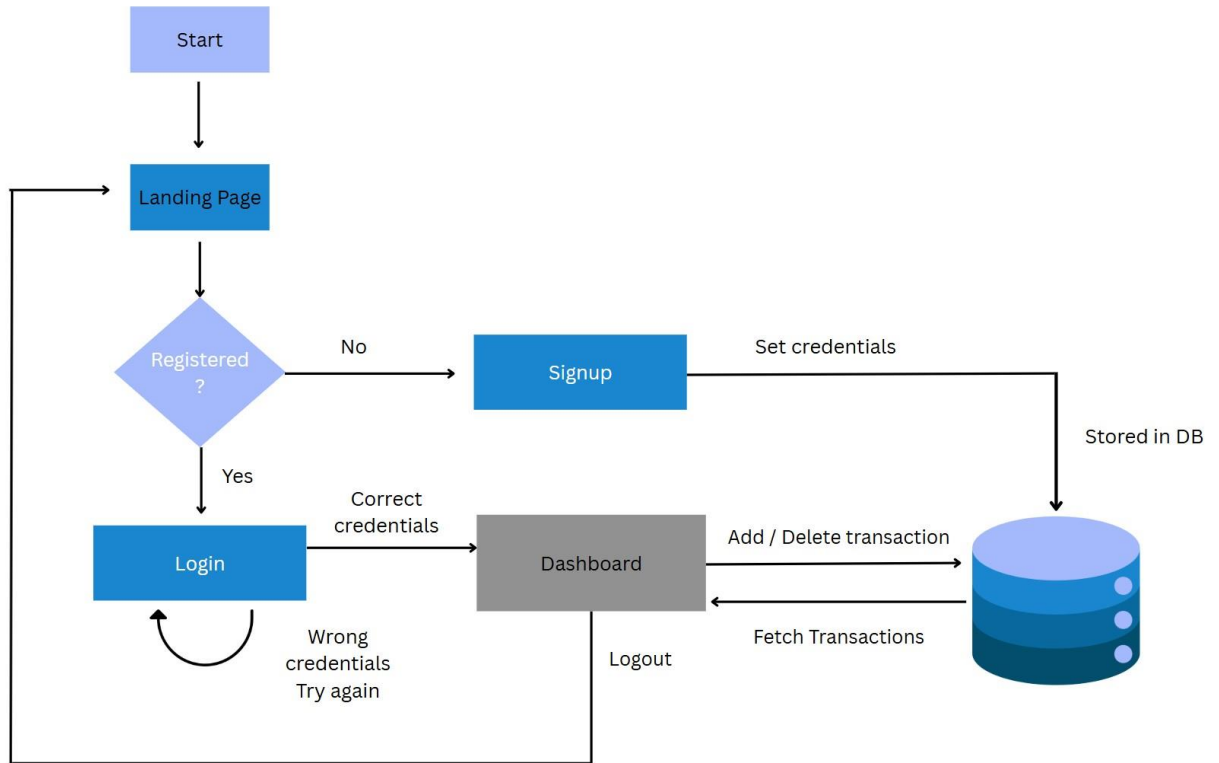
Akash — A 28-year-old working professional who wants to track his spending and save more each month.

	Journey Phase	Example: "Onboarding"	Exploring Features	Adding Expenses	Analysing Budget
	User Actions	Downloads app, signs up, sets budget preferences	Navigates through the dashboard and settings	Enters daily expenses, categorizes them	Views monthly reports and insights
	Touchpoints	App store, welcome screens, email confirmation	Dashboard interface, tutorial pop-ups	Add Expense" button, category list	Charts, insights screen
	Sentiments	This looks simple to set up	There are a lot of options; feels powerful	This is easy to update daily.	I didn't realize I spent this much
					
	Possible Solutions (Opportunities)	Add tooltips to guide through setup	Provide a quick tour or checklist	Enable voice input or receipt scanning	Offer personalized savings tips and recommendations

3.2 Technology Stack



3.3 Data Flow Diagram



4. PROJECT DESIGN

4.1 Problem-Solution Fit

- **Problem:**

Many individuals struggle to consistently track their daily expenses, leading to difficulties in budgeting, potential overspending, and a general lack of financial awareness. Traditional methods like spreadsheets or manual notes are often time-consuming, lack real-time feedback, and make it difficult to visualize spending patterns effectively.

- **Solution:**

The **MERN Personal Expense Tracker** provides a seamless, full-stack web application designed for efficient personal finance management. With features like secure user authentication (JWT & bcrypt), an intuitive dashboard displaying expense summaries, detailed expense/budget tracking (CRUD operations), integrated budget monitoring with visual progress and alerts, and on-demand reporting capabilities (spending by category, budget vs. actuals), the platform empowers users. It enables them to easily monitor spending habits, understand their cash flow through insightful reports (formatted in INR), and make informed budgeting decisions—all from one centralized, user-friendly, and responsive interface.

4.2 Proposed Solution

The **MERN Personal Expense Tracker** application implements a secure and modular full-stack architecture using the MERN stack (MongoDB, Express.js, React.js, Node.js) to manage personal finances effectively.

- **Authentication using JWT:**

Users initiate an authentication request (signup or login) by sending credentials to the backend Express.js server. Upon successful verification (checking hashed passwords with bcryptjs), the backend generates and issues a JSON Web Token (JWT) containing the user's ID. This token is securely signed using a secret key and sent back to the client (React), which stores it (typically in localStorage).

- **Protected Routes:**

For subsequent requests to protected resources (like the main Dashboard, Budget management page, Reports page, or user-specific data API

endpoints), the client includes the JWT in the Authorization: Bearer <token> header. Backend middleware (authMiddleware.js) intercepts these requests, verifies the token's validity, and fetches the associated user data, granting access only to authenticated and authorized users.

- **Data Flow:**

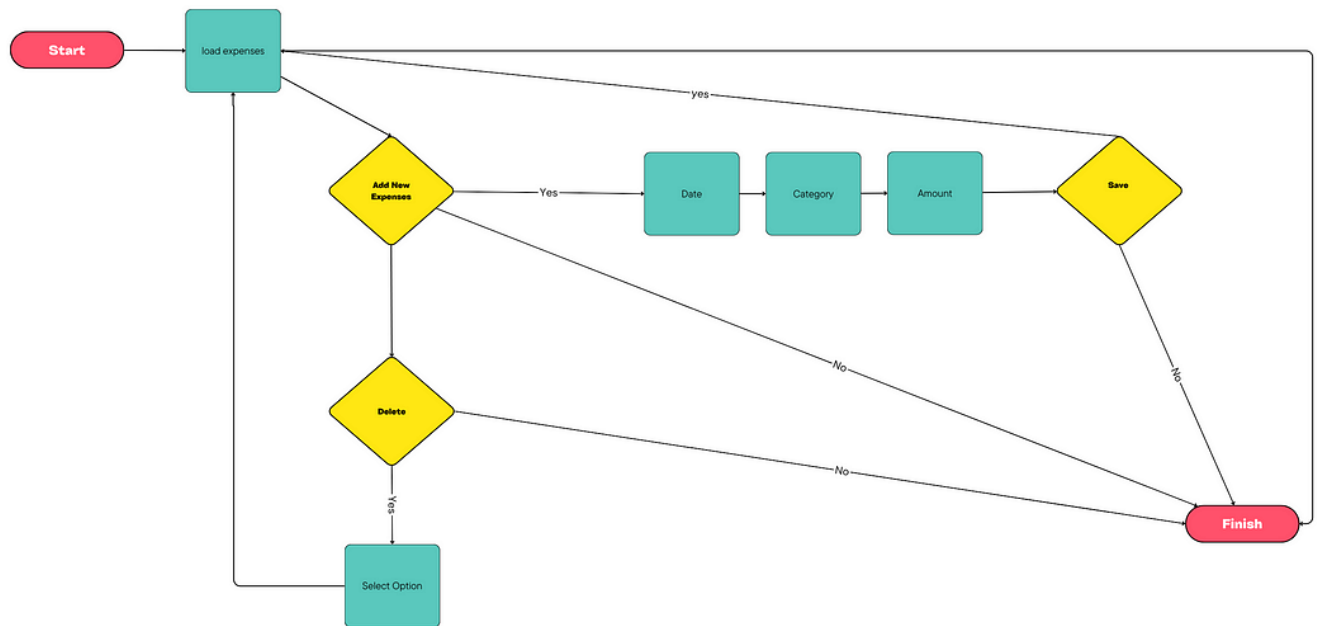
The frontend (React) application, structured with components, pages, services, and context for state management, interacts with the backend (Express.js) through a defined set of RESTful API endpoints. User actions like adding, viewing, updating, or deleting expenses and budgets trigger asynchronous API calls (via Axios) to the corresponding secure backend routes. The backend controllers process these requests, interact with the database, and send back appropriate JSON responses.

- **Database Integration:**

The backend utilizes **MongoDB** as its database, managed through the **Mongoose** Object Data Modeling (ODM) library. Mongoose defines schemas (User, Expense, Budget, Notification) which enforce data structure and validation rules. All user information, financial records (expenses, budgets), and notifications are stored persistently in MongoDB collections. Controllers use Mongoose methods to perform CRUD operations and complex queries (including aggregations for reports) against the database.

4.3 Solution Architecture

Expenses Tracker



5. PROJECT PLANNING & SCHEDULING

Sprint	Day(s)	Tasks	Deliverables
Sprint 1 – Setup & Authentication	Day 1–2	- Initialize MERN stack codebase - Set up MongoDB connection - Implement JWT-based login/signup - Basic UI for Authentication	Working JWT auth MongoDB schema for users Login/Register forms
Sprint 2 – Expense Tracking	Day 3–5	- Created Expense schema - Build CRUD APIs for expenses - Develop React components for add/view/delete - Protect APIs with JWT middleware	Add/View/Delete expenses Auth-protected routes Connected front-end

Sprint 3 – Dashboard & Deployment	Day 6–8	- Build expense summary dashboard (category) - UI improvements - Bug fixes - Final testing - Financial Reports - Business Receipts Upload	Functional dashboard Final UI completion
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6. FUNCTIONAL AND PERFORMANCE TESTING

S.No.	Parameter	Values	Screenshot
1	Application Summary	Personal-Expense-Tracker is a MERN-based expense tracker with user auth, expense management, and data visualization.	Signup page (Postman) Screenshot below
2	Authentication Functionality	-api/auth/register registers new users successfully. -api/auth/login returns JWT token on valid credentials.	Login response with token Screenshot below

3	Transaction Fetch	- <code>api/expenses</code> retrieves user expenses if JWT is valid. - Returns 401 error when the token is missing/invalid (expected behavior).	Fetch expenses
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The screenshot shows the Postman interface with a POST request to `http://localhost:5001/api/auth/register`. The request body is a JSON object:

```
{
  "name": "Postman Test User",
  "email": "postman@example.com",
  "password": "testpassword"
}
```

The response is a JSON object with the following structure:

```
{
  "success": true,
  "token": "eyJhbGciOiJIUzI1NiIsInR5cCI6IkpXVCJ9.eyJpZCI6IjY4MDA2ZTYyNTkxOGQ3NTZmOGIwMTg0NiIsIm1hdCI6MTc0NDg1MDcyMiwiZXhwIjoxNzQ3NDUwNzIyZGQ3ZWZmKRAzGFzG5-IDF21quDU6NZManXadWKFbUBGU0",
  "user": {
    "id": "68006e625918d756f8b01846",
    "name": "Postman Test User",
    "email": "postman@example.com"
  }
}
```

The screenshot shows the Postman interface with a POST request to `http://localhost:5001/api/auth/login`. The request body is a JSON object:

```
{
  "email": "postman@example.com",
  "password": "testpassword"
}
```

The response is a JSON object with the following structure:

```
{
  "success": true,
  "token": "eyJhbGciOiJIUzI1NiIsInR5cCI6IkpXVCJ9.eyJpZCI6IjY4MDA2ZTYyNTkxOGQ3NTZmOGIwMTg0NiIsIm1hdCI6MTc0NDg1MDgyNiwiZXhwIjoxNzQ3NDUwODI2fQ.cwRJaJh-p1YtF91BAC5E1y3rDN9QW511muXS--ahAQI",
  "user": {
    "id": "68006e625918d756f8b01846",
    "name": "Postman Test User",
    "email": "postman@example.com"
  }
}
```

7. RESULTS

Login Page:

Personal Expense Tracker

Take control of your finances, effortlessly.



Welcome Back!

Login

Don't have an account? [Register Now](#)

Dashboard Page:

Expense Tracker

Budgets

Reports

99

Dashboard

Welcome, Vikesh Mehta! Logout

Add New Expense

Total Recorded Expenses: ₹1,09,999.99

8 Apr 2025

Grocery (General Store)

Groceries

₹2,000.00

👍🗑️

8 Apr 2025

Clothing (Clothing Store)

Clothing

₹3,000.00

👍🗑️

8 Apr 2025

Went for movie night. (PVR)

Entertainment

₹1,499.99

👍🗑️

8 Apr 2025

College fees. (VIT University)

Education

₹1,00,000.00

👍🗑️

4 Apr 2025

I have went for buying clothes for me. (Monte Carlo)

Clothing

₹2,000.00

👍🗑️

3 Apr 2025

Grocery (General Store)

Groceries

Reimbursable

₹1,500.00

👍🗑️

Budget Page:

Expense Tracker

Dashboard

Reports

Manage Budgets

Set New Budget

Category* *

Budget Amount (INR)* *

Groceries

e.g., 10000

Period* *

Period Start Date* *

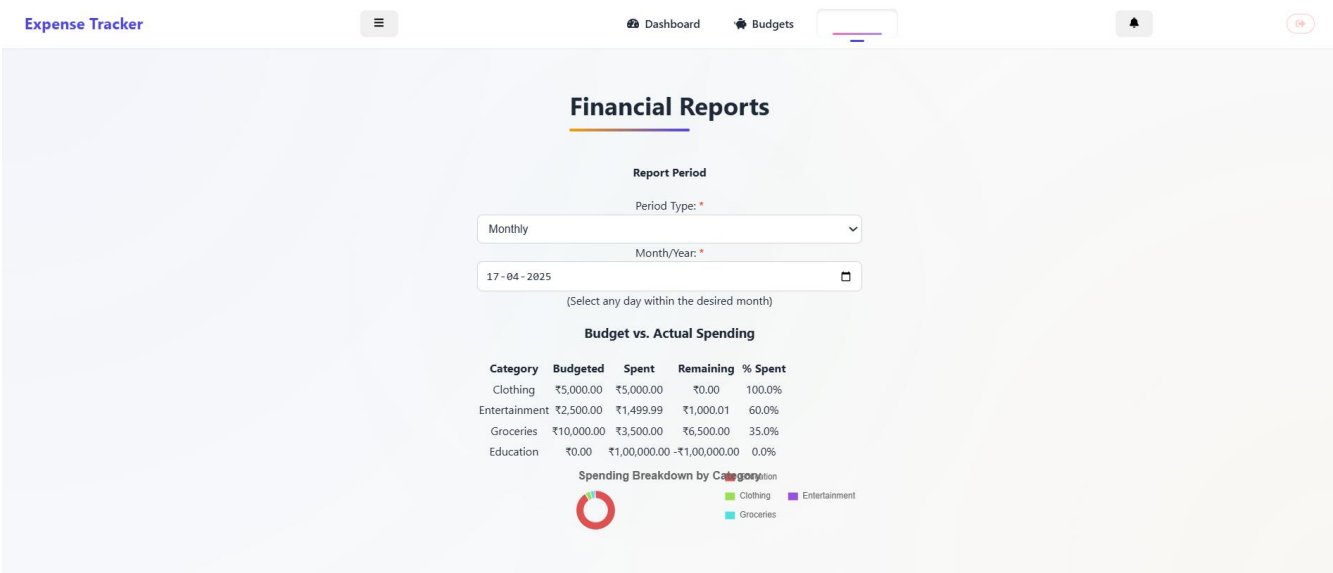
Monthly

17-04-2025

Notes

Set Budget

Reports Page:



8. ADVANTAGES & DISADVANTAGES

Advantages:

1. Comprehensive Tracking & Budgeting: Allows users to efficiently log (CRUD), categorize, and track expenses. Supports setting budgets across various periods (Monthly, Quarterly, Yearly, Weekly) per category, with visual progress tracking.
2. Actionable Insights & Reporting: Provides clear reports on spending habits (breakdown by category via charts) and budget performance (Budget vs. Actual

table), enabling users to analyze patterns and make informed financial decisions. Filters allow viewing data for specific periods.

3. **Clean and Intuitive UI/UX:** Features a user-friendly interface with clear navigation (Navbar), animated elements (Framer Motion), and consistent styling, aiming for ease of use and reducing clutter.
4. **Budget Alerts & Notifications:** Includes a system (backend checks, frontend polling & display) to notify users when they are approaching or have exceeded their set budget limits for specific categories.
5. **Secure Authentication:** Implements robust user authentication using JWT and password hashing (bcryptjs) to protect user data and sessions.
6. **Low Learning Curve:** Designed with standard web application patterns, making it relatively easy for users familiar with web apps to navigate and utilize its features.
7. **INR Currency Support:** Explicitly handles and displays monetary values in Indian Rupees (₹).

Disadvantages:

1. **Manual Data Entry:** Requires users to manually input all expense transactions. Lacks automation features like bank account linking or receipt scanning (OCR).
2. **Basic Notification System:** The current notification system relies on frontend polling at intervals, meaning budget alerts are not delivered in absolute real-time.
3. **Limited Business/Multi-User Features:** Primarily designed for individual or shared single-account use (like a family sharing credentials). Does not support distinct employee accounts, user roles, or advanced business reporting needed for robust business expense management.
4. **No Receipt Upload Functionality:** While a field exists for a receipt URL and a visual file input, the application does not implement actual file upload, storage, or management.
5. **Fixed Category List (UI):** Users cannot currently add, edit, or delete expense/budget categories directly through the user interface; the list is predefined in the backend code.
6. **Limited Offline Capability:** As a web application, it requires an active internet connection to interact with the backend and fetch/save data.

9. CONCLUSION

The Personal-Expense-Tracker application was developed with simplicity and speed in mind, targeting users who want a lightweight tool to manage daily finances. By skipping traditional expense categories, the app removes complexity and focuses solely on helping users keep track of how much they're spending vs. earning.

It addresses the need for a no-frills expense manager—ideal for users overwhelmed by feature-heavy alternatives. Though currently minimal in scope, the app successfully delivers on its primary goal: efficient and straightforward personal finance tracking.

10. FUTURE SCOPE

- Add Optional Categorization: Include optional tags or categories without making them mandatory.
- Automated Data Entry: Integrate with SMS/email parsing or bank APIs for automatic transaction logging.
- Customizable Reports: Enable users to generate visual summaries and downloadable reports.
- Expense Reminders: Introduce gentle nudges or notifications to keep users consistent in logging.
- Dark Mode & UI Personalization: Enhance accessibility and improve user experience.
- Multi-user Support: Add shared wallets or group tracking features for families and roommates.

GitHub & Project Demo Link:

GitHub: https://github.com/Vikesh-Mehta/PERSONAL_EXPENSE_TRACKER_APP

Demo Link: https://drive.google.com/file/d/1Df84UKj0Wyak9r_Im7ziqKgQbP_irR80/view?usp=sharing